

Summarized chain of thought (Kaplansky Direct Finiteness in Characteristic Two)

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The central algebraic question is whether $ab = 1$ forces $ba = 1$ in every positive-characteristic group algebra. Part I develops a proposed characteristic-two counterexample. Part II shows how finite-field witnesses $ab = 1, ba \neq 1$ give an injective, nonsurjective cellular automaton. That conversion assumes the witnesses furnished by the previous characteristic-two argument.

Part I: Positive-characteristic direct finiteness

Original prompt (excerpts)

Problem

Resolve the following unrestricted positive-characteristic form of Kaplansky's direct-finiteness conjecture, by a complete proof or a rigorous counterexample.

TARGET AND DEFINITIONS

For every prime p , every field K of characteristic p , every group G , and every a, b in the group algebra $K[G]$, prove

$$ab = 1 \text{ implies } ba = 1.$$

Here $K[G]$ consists of finite formal sums $a = \sum_g a_g g$ with a_g in K , with convolution multiplication induced by the multiplication in G ; its identity is the basis element corresponding to the group identity. Groups with torsion are included. Neither K nor G is assumed finite. It is enough to establish the assertion for every finitely generated G and then explain the reduction: the union of the finite supports of a and b generates a subgroup H , and $K[H]$ embeds in $K[G]$.

The target is direct finiteness of group algebras, not the assertion that every group has no zero divisors or only trivial units. In particular, proving a zero-divisor theorem only for torsion-free groups does not resolve this task. Stable finiteness asks the analogous identity $AB=I$ implies $BA=I$ for every finite square matrix size. That stronger conclusion is welcome if proved, but distinguish it from direct finiteness and justify every claimed equivalence; direct finiteness does not imply stable finiteness for arbitrary rings.

RESEARCH DIRECTION

Develop a general argument or an explicit obstruction. Existing proofs for characteristic zero and for surjunctive groups are useful starting points. Any proof restricted to sofic, residually finite, amenable, or surjunctive groups must identify its restriction instead of silently extending it to arbitrary groups. Do not assume all groups satisfy one of those approximation properties. A theorem about complex operator algebras cannot be transferred to characteristic p without a valid argument accounting for that change of coefficients. If specializing finitely generated coefficient rings to finite fields, preserve the equation $ab=1$ and the nonvanishing of any coefficient witnessing $ba=1$, and prove the required specialization exists.

A useful first milestone is a new class of groups not covered by the known sufficient hypotheses, or a precise local-to-global criterion whose remaining hypothesis is explicitly isolated. These milestones do not replace the full quantifiers of the target.

Check explicitly where finite support, the field characteristic, torsion, and any approximation assumption enter. Verify the argument recovers the elementary finite-group case, where $K[G]$ is a finite-dimensional K -algebra even when its characteristic divides the group order.

For a counterexample, specify a prime, a field with an exact representation, a well-defined group, and two finite sums. Certify the multiplication $ab=1$ and exhibit a nonzero coefficient or other rigorous certificate for $ba=1$. A finitely presented group together with unproved word equalities or inequalities is not a certificate. One-sided shifts in an unrestricted infinite-dimensional endomorphism ring or an example in a nongroup monoid algebra do not qualify.

STARTING PRIMARY REFERENCES

1. Tullio Ceccherini-Silberstein, Michel Coornaert, and Xuan Kien Phung, Stable finiteness of monoid algebras and surjunctivity: <https://arxiv.org/abs/2405.18287> ; published version <https://doi.org/10.1016/j.tcs.2025.115228> . Provides a known sufficient surjunctivity hypothesis, not an all-groups solution.
2. The same authors, First-order model theory and Kaplansky's stable finiteness conjecture for surjunctive groups: <https://ems.press/content/serial-article-files/51166?nt=1> . Gives precise ring definitions and a model-theoretic proof under surjunctivity. Its historical comments about nonscific groups must not be assumed current.

1 From candidate dimensions to characteristic-two sign lifting

The investigation sought direct finiteness, $ab = 1 \Rightarrow ba = 1$, for arbitrary positive-characteristic group algebras. The assistant used the supplied surjunctivity results to identify a sufficient hypothesis for stable finiteness [12, 13]. It also recorded Elek and Szabó's sofic-group theorem as a known special case [19]. The idempotent defect $q = 1 - ba$, with $q^2 = q$ and $aq = qb = 0$, organized its search beyond that hypothesis.

Witt-vector lifting suggested a zero-trace idempotent, but the assistant could not supply the positivity or continuation needed to force vanishing. Its entropy approach required strict decrease on proper linear subshifts; nonamenability obstructed naive entropy. It also used the universal left-invertible-element groups of Dykema, Heister, and Juschenko to reduce the problem to finite cancellation patterns [17].

Random-priority pivot densities suggested an algebraic dimension, but the proposed conservation argument required differentiating infinite-volume quantities across possible phase transitions. The assistant did not establish this step. Enlarging coefficients to a skew group ring instead shifted the difficulty to an unresolved rank condition.

Spectral amplification sought a forbidden rational spectral atom from repeated defect summands. Its gains remained conditional on coefficient, support, denominator, and determinant control; the assistant connected this route with Thom's sofic eigenvalue and integer-operator work [37, 38] and the sofic determinant result discussed by Balci and Skandalis [4]. It still needed an obstruction valid for arbitrary groups.

The final turn specialized to characteristic two. Pairing cells with equal products $x_i y_j = x_k y_l$ suggested replacing these relations by $x_i y_j = z x_k y_l$, where a central involution z would supply minus signs in a complex representation. Success required z to survive. Reduced row cycles yielded reduced column words, which encouraged the assistant, but candidate odd-area spherical diagrams defeated the simple parity justification. Their completion to a full cancellation pattern remained unresolved.

VERBATIM EXCERPT

Need show there is always *some* pairing enabling sign lift, or refine by grouping classes size>2 with arbitrary phases. However if equalities exactly 2, no alternative.

In considering whether characteristic-two units could be lifted, it recalled Gardam's nontrivial units in characteristic two and over the complex numbers [20, 21]; it did not obtain the lifting principle it wanted. The assistant ended by reformulating the row graph as an immersion into a graph with column-labeled edges, leaving the required group-theoretic control unresolved.

2 Positivity barriers and asymmetric cancellation

The same attempt moved between proving direct finiteness, $ab = 1 \Rightarrow ba = 1$ in positive-characteristic group algebras, and constructing a counterexample. Known reductions connected the universal problem to finite coefficient fields and surjectivity [12, 13]. In particular, the assistant read the Dykema–Juschenko reduction equating stable finiteness of $K[G]$ with direct finiteness of $K[G \times H]$ for every finite group H [18]. It also continued using universal left-invertible-element presentations to encode characteristic-two cancellations [17].

Random valuations first sought a substitute for positive Hilbert-space geometry. Distinct leading weights prevented finite cancellations, but infinite elimination could fail to converge, and coercivity did not establish surjectivity. A separate argument developed within this attempt amplified a hypothetical defect through lamp projections and tensor powers. It proposed integral positive operators with negative logarithmic determinant, conditional on Lück’s determinant conjecture for the relevant product groups. Positivity and moment integrality did not remove that hypothesis.

Manzoor’s work separated the determinant conjecture from co-hyperlinearity for invariant random subgroups; it did not remove the group hypothesis in the proposed amplification [28].

Characteristic-two cancellation then suggested comparing right and left quotients of finite supports. No general parity transfer emerged. A narrower argument observed that collisions under conjugation occur at isolated indices or periodically. It proposed that an idempotent e satisfying $(1 - e)te = 0$ for a group element t must also satisfy $(1 - e)t^{-1}e = 0$.

This obstructed compression by a single group element; extending it to polynomial sums remained open. Finite-subgroup averages next reduced a possible counterexample to asymmetric double-coset congruences. Incidence designs promised forward cancellation, but curvature constraints and collapsing presentations blocked the attempted constructions. The assistant still needed to show that an odd reversed class survived the imposed relations.

Finally, polynomial matrices over the one-sided algebra $k\langle x, y \mid xy = 1 \rangle$, already considered within the attempt, seemed to lift forward Laurent identities while preserving a reversed defect. Constant-intersection restrictions blocked the simplest construction. Split Leavitt families and singular matrices left a further possibility, but the closing word-count argument was unfinished. This stage ended without a proof or counterexample.

3 Coset constructions and the search for positive dimension

The assistant’s search for a failure of direct finiteness in characteristic two gradually gave way to attempts to prove it through dimension. The target remained the implication $AB = 1 \Rightarrow BA = 1$ in a group algebra.

A corrected rank argument handled a special linear construction with split relations, without extending to arbitrary group-algebra elements. The construction search then sought odd-order finite subgroups whose nonidentity elements cancel with even multiplicity. Their averaging projectors could provide a rectangular factorization; converting it into a counterexample required faithful subgroup embeddings and compatible character projections. Dense identifications threatened collapse, while high-girth constructions exceeded the required size bound.

Recasting the subgroup identifications as linear-system solution groups led the assistant to Slofstra’s embedding theorem and its wagon-wheel construction [35], as a possible way to retain elements invisible to finite-dimensional representations. It also considered whether the machinery of $\text{MIP}^* = \text{RE}$ [25] could produce the required compatible measurements.

VERBATIM EXCERPT

Cosets give much more flexibility! Any hyperedge set affine copies. Equality 6 mod p means symmetric difference of t odd-size cosets leaves singleton.

Translated subgroup indicators allowed parallel affine pieces, but realizing their cancellation while keeping $\sum_i |K_i|^{-1} \leq 1$ remained unresolved. Curvature estimates and a character-based intersection obstruction constrained particular constructions without excluding all possibilities.

The positive strategy sought a bounded, strictly monotone dimension invariant under finite-propagation changes of basis. The assistant had not justified the required conservation identity for independent random pivot orders. Averaging orders lacked finite normalization, and a quantum mutual-information transport faced unattained pivot thresholds and conditioning problems.

Before returning to algebraic constructions, the assistant recalled the discussion in Dykema, Heister, and Juschenko [17] connecting the universal direct-finiteness and stable-finiteness problems, and considered transferring a matrix failure to a scalar failure over a larger group.

Other routes exposed related limitations. A sparse-orthogonality argument required uniformly bounded supports, whereas the defect-generated families grew. The group-theoretic matrix-multiplication approach of Cohn and Umans [14] suggested seeking tensor-rank bounds on the proposed matrix units, but the assistant encountered the absence of a finite ambient volume. Looking at Manzoor's work on invariant random subgroups satisfying the determinant conjecture [28] did not supply a theorem for arbitrary groups. Finite determinant estimates lacked a justified infinite passage. The later Krylov-space argument suggested an analogy with the information-flow index of Gross, Nesme, Vogts, and Werner [23]; the assistant sought a corresponding index for its time evolution, while nonamenable support expansion still prevented the dimension-growth estimate from yielding a contradiction.

VERBATIM EXCERPT

Formal p-adic solution with support growing not implies finite algebraic solution; complete filtered ring maybe algebraic power series coordinates with infinite group monomials.

Thus coefficient lifting also failed to preserve the required finite support. The assistant ended this stage without a coset counterexample or a universal positive dimension, leaving its final module-localization proposals conditional.

4 From modular cancellation to restricted incidence obstructions

The search for $ab = 1$ but $ba \neq 1$ in a group algebra moved between algebraic constructions and analytic obstructions. The universal left-invertible-element presentations of Dykema, Heister, and Juschenko [17] framed the search for finite models of the forward product relations. Laurent-series completion lacked the flatness needed to preserve an injection; characteristic-two phase lifts required unproved global consistency.

The assistant recalled the known sofic and surjunctive cases [19, 12, 13] while asking whether finite models or logical compactness could go beyond those hypotheses. It did not obtain such an extension.

The attempt's own subgroup-covering proposal required t actual elementary abelian subgroups K_i , each of odd order k , with t odd and $2 \leq t \leq k$. Even multiplicity of every nonidentity element would give

$$\sum_i \widehat{K}_i = 1 \quad \text{in } \mathbb{F}_2[N], \quad \widehat{K}_i = \sum_{g \in K_i} g.$$

A proposed conversion to a one-sided inverse remained conditional on faithful subgroup realization.

Meanwhile, the rational-eigenvalue route was related to Thom's work on sofic groups and integral operators [37]. Tensor amplification suggested a determinant obstruction, but the assistant found that integer moments allowed small exceptional spectral mass to escape toward large values; the required positivity principle was missing. The subgroup-covering search also recalled Arkhipov's connection between graph planarity and binary magic games [3]. The assistant asked whether analogous noncommutative flows could retain an odd-order defect; it did not establish that extension.

The construction then relaxed complete cancellation to a cylinder set E in a probability-preserving lamp action. Its indicator compressed subgroup sums, with disjoint translates removing uncanceled terms. The desired density bound was $t/k \leq \mu(E)$. An invariant-matching approach recalled Lyons and Nazarov's result for bipartite Cayley graphs of nonamenable groups [27]; a matching principle for the more general incidence system remained missing. Associated active blocks formed finite linear spaces. The assistant used the de Bruijn–Erdős finite linear-space bound [11], in its Fisher-type argument, to force the averaged reciprocal-size cost above one for a noncollinear block surrounded by leaves:

VERBATIM EXCERPT

Excellent. This kills path/star if neighbors always leaves. If neighbors internal, cost shared; each line may participate in growing network, maybe use Euler characteristic.

Interacting blocks, odd cycles, and realization of the cylinder remained unresolved. The assistant ended with this restricted obstruction and no completed construction or proof of unrestricted direct finiteness.

5 Restricted obstructions and the limits of local information

The assistant investigated the characteristic-two case of direct finiteness through the stronger stable-finiteness question: whether $AB = I$ forces $BA = I$ for square group-algebra matrices. A hypothetical failure supplies the nonzero idempotent $q = I - BA$.

In considering a compact-shift formulation, it recalled Seward's equal-entropy isomorphism theorem for Bernoulli shifts over countably infinite groups, while asking what could distinguish the proposed product decomposition [33]. The assistant recalled the Andrásfai–Erdős–Sós theorem that a triangle-free graph with minimum degree greater than $2n/5$ is bipartite [1], and Slofstra's embedding theorem for finitely presented groups in solution groups [35].

The next direction treated a split counterexample as a local isomorphism $X \cong Y \times Z$, where X, Y are equal-alphabet full shifts and Z is nontrivial. A biased independent-input path suggested the entropy differential $dS = -\langle \phi_t, d\mu_t \rangle$. The assistant used finite dependence to argue for boundary cancellation along that path, then identified unresolved requirements: global path independence, positivity, and preservation of a finite-range Gibbs description when retaining only the complementary factor.

Information allocation and linear elimination then encountered unbounded propagation. The assistant drew an analogy with the information-flow index of Gross, Nesme, Vogts and Werner for one-dimensional quantum cellular automata, seeking a counterpart on nonamenable groups [23]. A sharper circuit proposal used

$$W = \begin{pmatrix} B & q \\ 0 & A \end{pmatrix},$$

whose inverse and elementary-shear decomposition follow from the split relations. Apparent compression of one logical layer motivated an entropy-balance invariant. The assistant constructed a free-group controlled-phase graph state and concluded that it defeated the broader premise:

VERBATIM EXCERPT

| So equality of entropy densities for pure finite-depth states is false. Check: C inject from degree2 to first via boundary operator on free F2, no compact kernel (tree), yes.

VERBATIM EXCERPT

| Key EXTRA: in AB case, the entanglement of each S site has Bell *local purification in T*.

The assistant retained bounded-neighborhood Bell recovery as a possible extra hypothesis, but did not find an invariant using it. Arithmetic lifting finally returned to the same locality difficulty: formal corrections could enlarge support indefinitely, so the assistant had not obtained a finite-support characteristic-zero inverse.

6 Obstructions to sign lifting and modular corner constructions

The attempt continued seeking direct finiteness: whether $ab = 1$ forces $ba = 1$ in arbitrary positive-characteristic group algebras. The assistant retained surjunctivity as the sufficient hypothesis in the two starting papers of Ceccherini-Silberstein, Coornaert and Phung [12, 13], and recalled Elek and Szabó's sofic case [19]. An integral-lift approach associated a divisible module to a hypothetical defect. The assistant used a shear-and-dilation example to argue that finite generation permits unbounded primary torsion; the missing constraints were projectivity and a square presentation. Decay of inverse coefficients likewise failed to localize the defect.

Returning to the universal left-invertible-element presentations of Dykema, Heister and Juschenko [17], the assistant shifted to lifting characteristic-two cancellation pairs to opposite signs in characteristic zero. Earlier reasoning within this same attempt supplied the obstruction: odd spherical diagrams can collapse the required central involution. It found that tensor pairings preserved that obstruction.

Cancelling at the first unsuccessful position in a tensor product appeared cheaper, but the assistant calculated that correction ranks consumed the entire proposed saving. A compact/discrete duality argument then lacked a well-defined preserved pairing.

The assistant also revisited its conditional spectral approach, relating a hypothetical modular defect to a nonintegral rational eigenvalue of an integral group-ring matrix. Thom's work on sofic groups and integer operators supplied the spectral comparison [37]; the assistant still needed a determinant lower bound for the unrestricted groups under consideration.

Cyclic cancellation paths retained only conjugacy-class information:

VERBATIM EXCERPT

| This yields Hattori trace over conjugacy class as expected. Want individual h not sum.

7 Modular decoding meets information at infinity

The positive-characteristic direct-finiteness attempt, seeking $ab = 1 \Rightarrow ba = 1$, moved from small algebraic configurations to an information-theoretic obstruction. Characteristic-two calculations with two copies of S_3 led the assistant to exclude a small corner construction through double-coset analysis and finite or virtually abelian reductions. They did not address arbitrary groups.

It briefly considered returning to the ULIE presentation approach [17], then pursued the spectral route associated with Thom's work on sofic groups and algebraic eigenvalues [37]. The assistant returned to

its proposed reduction to an integer group-ring matrix $M = I + pC$ with normalized complex von Neumann rank $r < 1$. This suggested a tension: $MX \bmod p = X \bmod p$ recovers independent input digits locally, while low spectral rank suggests limited information. Tensor powers promised rank at most r^j , while their operator norms grew at most as K^j for a fixed K . The Ornstein–Weiss free-group gradient example [31] defeated a naive output-only argument, and carry dynamics could import information from outside every finite observed region. For the Gaussian channel $Y(t) = \sqrt{t}MX + Z$, with independent standard Gaussian noise Z , the assistant compared local digit recovery with a spectral upper bound involving $\frac{1}{2}\tau \log(I + t\sigma^2 MM^*)$, where σ^2 is the input-digit variance and τ is normalized trace. It used the finite I-MMSE identity [24] to seek a lower bound from local decoding, hoping tensor amplification would force a contradiction:

VERBATIM EXCERPT

| This would be a full proof if derivative identity (no lost information to infinity) works.

The infinite-output limit gave an inequality in the wrong direction. A proposed random-order Gaussian prediction identity did not solve the problem, because real predictor coefficients need not preserve modular decoding.

Reconsidering the attempt’s own tree-code example reversed a suspected entropy-defect sign. The next approach used systematic encodings $Y_i = (X_i, E_i)$, with independent inputs X_i and linear checks E_i . Under independent random site priorities, the assistant sought to match input information already revealed by earlier checks to information newly contributed by other checks. Finite elimination appeared promising:

VERBATIM EXCERPT

| Since only finitely many rows hit S , descending elimination terminates (ignoring rows zero on S), regardless chains through others! Excellent.

But finite matching conditions did not yield an invariant density comparison, and decoding thresholds might never be attained. Smooth score functions failed to capture discrete digit identity; the final lattice-divisibility approach invoked Bombieri–Vaaler bounds on small integer kernel vectors and their gcd-of-minors refinement [8], but faced uncontrolled boundary growth. The assistant ended with candidate information lower bounds whose passage from finite decoding to invariant averages, and whose control of expanding boundaries, it had not resolved.

8 Information arriving from infinity

The characteristic-two direct-finiteness attempt sought to contradict a nonzero defect $I - BA$ when $AB = I$. A reduction developed earlier within this same attempt proposed excluding integer finite-propagation operators $M \equiv I \pmod{p}$ with positive-dimensional real kernel. Independent base- p digits remain exactly recoverable from MX modulo p , whereas amplification was intended to make its von Neumann rank small. The assistant sought a comparison between local recoverability and global information capacity to complete the contradiction.

The assistant tried arithmetic determinants, torus dynamics, and determinantal selections of kernel coordinates, but found no such comparison. It observed that, on nonamenable groups, finite independence need not exclude an infinite square-summable kernel, while local maps can increase the number of independent information streams. It recalled Seward’s connection between positive Rokhlin entropy and the entropy of Bernoulli shifts, and worried that its proposed entropy inequality would require resolving that broader question [34]. These objections redirected its effort toward Gaussian observations and identities relating information to mean-square error.

VERBATIM EXCERPT

| True Brownian with static drift. Thus ****reverse**** process diffusion with known simple linear dynamics product across sites, independent! This is huge.

The enthusiasm concerned the independent Brownian bridges obtained by reversing the observations. The assistant tried to turn this structure into an information identity through a representation or uniqueness theorem, but did not establish either. Possible posterior transitions and information arriving from arbitrarily distant coordinates challenged its unrestricted form. A spin-flip calculation suggested a low-signal estimate, but higher-order terms obstructed extension.

The assistant then proposed a spatial Poincaré inequality, but recognized that it did not give the compactness needed for uniqueness:

VERBATIM EXCERPT

| Poincaré alone no compact embedding (infinitely many shifts e_i). $q_s \rightarrow 0$ could carry energy in changing spatially far modes with bounded gradient, not high temporal frequency.

Finite-coordinate approximation introduced covariance errors whose total energy might survive while escaping outward. High-signal disagreement bounds covered only a later regime, leaving the earlier information deficit uncontrolled.

The attempt then recast the algebraic compression using commuting local quantum matrix factors and Bell references. Each logical factor had a fixed outgoing mutual-information bound, but conditioning could let one physical site unlock information about many references, defeating the required incoming bound. It also considered an information-flow invariant analogous to the one-dimensional index of Gross, Nesme, Vogts, and Werner, but left its extension across the proposed nonamenable interface undefined [23]. High-temperature expansions and projective-unitary phases supplied further speculative directions. The assistant ended with both the Gaussian uniqueness argument and the quantum capacity argument unresolved.

9 From Gaussian rank bounds to vertical defect components

The attempt sought to connect arithmetic congruence with analytic rank in the unrestricted positive-characteristic direct-finiteness problem, where $ab = 1$ should imply $ba = 1$ in a group algebra. Its working obstruction was an integral operator $M \equiv I \pmod{p}$ with deficient von Neumann rank, chiefly for $p = 2$. Tensor powers would amplify the deficiency. This approach built on the earlier finite-subgroup and integer-flow proposals.

Periodic Gaussian measurements offered a possible bridge: congruence bounds signed Fourier moments, suggesting nearly independent binary observations. Yet their entropy could not be bounded directly by real Gaussian capacity, because conditioning on binary observations discards information retained by real measurements. A subsequent lattice theta estimate seemed less sensitive to the operator's support degree:

VERBATIM EXCERPT

| Excellent! No support degree! Let's verify Poisson/unimod lattice for coset shift.

The proposed bound controlled finite lattice sums, but the required infinite-volume determinant comparison remained missing. Wrapped Gaussian arguments encountered hidden winding numbers; finite-modulus inverses could send relevant modes to increasing frequencies and supports. The assistant

found that general Cameron–Martin and logarithmic Sobolev principles did not supply the needed comparison.

After analytic-continuation and torsion proposals stalled, the attempt turned to the fixed-support coefficient ring

$$A = \mathbb{Z}[a_s, b_t] / (\text{coefficients of } ab - 1).$$

For $q = 1 - ba$, it argued that characteristic-zero direct finiteness makes the coefficients of q torsion elements in the additive group of A . Their finitely generated ideal satisfies $J^2 = J$, so it is generated by an idempotent. The proposed conclusion places any defect on an open-and-closed component supported over finitely many primes.

Excluding that component remained the missing step. Hensel lifts did not suffice because their supports grow, whereas the annihilating integer belongs to one fixed-support scheme. The assistant therefore retained the vertical component as the obstruction its deformation and lifting proposals still had to exclude.

10 Trace, entropy, and boundary barriers to direct finiteness

The search for direct finiteness continued from $ab = 1$ toward excluding the defect $q = 1 - ba \neq 0$ in $\mathbb{F}_p[G]$, especially in characteristic two. The assistant extended constructions developed earlier in the attempt.

The assistant found that coefficient schemes suggested that defects might occupy components confined to positive characteristic; enlarging supports did not connect these components to ordinary units. Integral lifting likewise lost control of support or dimension. A distributional telescoping proposal sought $x - bxa = r$, where $r = qgq$ has nonzero coefficient trace $\tau(r)$. It rejected this proposal because cyclicity forces $\tau(x - bxa) = 0$: compactness retained trace-bearing tails instead of producing the required cancellation.

The assistant also reconsidered the ULIE presentations of Dykema, Heister, and Juschenko, asking whether the dense short relations forced sufficient residual or geometric structure to settle reverse multiplication [17]. This produced no new criterion.

Attention shifted to locally split linear encodings of independent physical variables into logical variables. Ferromagnetic biases and mass transport seemed capable of bounding logical density by physical density:

VERBATIM EXCERPT

This yields monotonicity and ****pressure path inequalities via MTP**** maybe enough without amenability! New promising. Analyze carefully.

The assistant identified the missing step as passage from finite-volume mixed-response symmetry to infinite volume. Phase transitions could carry response beyond finite neighborhoods. Pinning finitely many physical coordinates recovered only the ordinary finite-support bound. Local stochastic dynamics did not repair the gap: the Ornstein–Weiss gradient example [31] showed that proposed entropy currents need not depend only on endpoints. Local splitting remained a stronger, unexploited condition.

A subsequent lattice approach used an earlier provisional reduction to an integer matrix $M \equiv I \pmod{p}$ with nonzero ℓ^2 kernel. Bounded integer solutions must vanish by repeated divisibility. The assistant invoked the Bombieri–Vaaler form of Siegel’s lemma to bound the product of sizes of independent integer solutions of finite subsystems [8]. Those short solutions could nevertheless concentrate near the boundary; nonamenable growth prevented the needed interior conclusion.

Finally, finite-subgroup sums and rank-one matrix weights sought a modular identity $\sum_i \widehat{K}_i W_i = I_m$ at low dimension cost. Higher-multiplicity identifications offered coding freedom, while cross terms imposed new constraints:

VERBATIM EXCERPT

| Offdiag conditions impose additional square density, likely second link counts.

The assistant ended without establishing the subgroup embeddings or the required identity, leaving the trace, entropy, and lattice approaches incomplete.

11 Matrix cancellation and punctured affine geometry

The attempt sought a characteristic-two counterexample to direct finiteness by combining matrix cancellation with a high-girth hypergraph. It sought local cancellation of rank-one matrix weights with a large nonzero global sum, followed by a group-algebra construction.

For t vertices and degree D , rank-one matrices W_v should sum to zero on every hyperedge, while their total has rank m large enough that $t \leq m(D-1)^2$. Incidence girth at least twelve was intended to support subgroup embeddings. A corrected size bound worsened the scalar size bound; polynomial evaluation examples defeated a proposed minimum-support rank inequality, leaving sparse local constraints as the central difficulty. A speculative detour invoked $\text{MIP}^* = \text{RE}$ while asking about solution-group encodings [25]. It also considered design-matrix rank bounds of Barak and collaborators, noting the difficulty of obtaining the needed finite-field bound [5]. Layered basis changes appeared to force short cycles. A later quantum-code reformulation considered the Bravyi–Poulin–Terhal tradeoff, but the assistant regarded geometric locality as unsuitable for the high-dimensional setting [9].

Descriptions of Golowich and Lin’s multiplication-based quantum LDPC construction and Dinur, Liu, and Zhang’s affine-line codes entered the search for local polynomial constraints [22, 15]. The assistant later acknowledged their possible inspiration, while continuing to develop the line construction itself. The next proposal removed holes O from $A = \mathbb{F}_q^n$, with q a power of two. Low-degree polynomial products sum to zero on full affine lines, whereas removing holes can produce a nonzero global matrix sum. Trades replacing radial lines through a hole by parallel lines avoiding it aimed to preserve regularity. Initial triangle estimates contradicted the dimensions needed for a local-lemma argument:

VERBATIM EXCERPT

| Need $b < a-1$ for rare. Generic LLL wanted $b > a+4$. Conflict!

A punctured-plane tiling offered a revision. For a linear map J with no eigenvector over $\mathbb{F}_q$, the line $\ell(x) = 1$ was lifted to $(x, \ell(Jx))$. Independence of x, Jx for $x \neq 0$ made these lifted lines partition every nonzero fiber. Sampling holes from a Newton interpolation information set would preserve independent evaluations after deletions. Interpolation polynomials then suggested

$$\sum_{v \in A \setminus O} W_v = -I_{|O|},$$

while retaining local line cancellation. Proposed cycle estimates appeared compatible with the rank bound.

The assistant then set out the remaining checks: cycle classification, probabilistic dependence bounds, subgroup embeddings, and conversion from a matrix obstruction to finite scalar group-algebra sums. It proposed attaching additive groups $\mathbb{F}_l^2$ to vertices, identifying one-dimensional subgroups along hyperedges, and using nonpositive curvature to establish embeddings. Subgroup averages were intended to

produce the matrix factorization, followed by character twists in HNN extensions and a finite-group matrix corner for the scalar passage. Linnik's theorem remained a possible dependency for choosing $l + 1$ divisible by q within polynomial bounds [26]. This stage ended with the assistant reconstructing the probabilistic hypotheses.

12 From a characteristic-two construction to a complete draft

The assistant moved from local audits to a complete candidate counterexample to direct finiteness, the implication $ab = 1 \Rightarrow ba = 1$ in a group algebra. It reconstructed the proposed characteristic-two construction from its preceding reasoning and used the supplied surjunctivity results as a consistency constraint: their stable-finiteness conclusions applied to surjunctive groups, a hypothesis it had not established for its proposed group [12, 13]. It likewise treated direct finiteness for sofic groups as a constraint on any proposed example [19].

The geometric proposal removed sparse interpolation points, called holes, from an affine space over $\mathbb{F}_q$, with q a power of two. Replacing lines near holes was intended to preserve regular degrees while excluding short incidence cycles. Random subspaces and local-lemma estimates supported this proposed existence argument; Linnik's theorem controlled the size of a prime $\ell \equiv -1 \pmod{q^{100}}$, giving $D = \ell + 1$ divisible by q [26]. Repeated scrutiny focused on whether repeated polygon labels really involved independent variables.

For m deleted interpolation points, polynomial value columns x_v at the remaining points were intended to satisfy $\sum_v x_v x_v^T = -I_m$, while their Gram sums on every retained line vanished. The assistant checked that products of the interpolation polynomials had degree at most $q - 2$, so their sums along full affine lines were zero. Finite groups attached to points converted these identities into rectangular matrices U, T with $UT = I_m$. An earlier route appealed to Haefliger's developability theorem for nonpositively curved complexes of groups, recalled through Bridson–Haefliger [10]; the assistant checked local links with girths 6, 4, 12 and triangle angles $\pi/3, \pi/2, \pi/6$. It then replaced this route with a proposed planar balanced-diagram argument using degree bounds, long faces, and Euler's formula for the local-group embeddings. With t remaining points, the bounds $m < t \leq m\ell^2$ allowed the proposed packing of subgroup averages into m slots by character twists and conjugating generators. The assistant argued that a reverse identity would, after applying augmentation in the embedded base group algebra, equate I_t with a matrix factoring through dimension m . An auxiliary finite-group matrix block was intended to yield two scalar group-algebra elements. The assistant related this step to Dykema–Heister–Juschenko's result that direct finiteness after taking products with every finite group implies stable finiteness of the original group algebra [17].

The assistant investigated possible rank bounds contradicting the polynomial construction and rechecked the embedding and packing steps. It completed the candidate draft, then returned to the probabilistic estimates. In particular, it argued that pairwise trivial intersections of direction spaces separated the cosets for repeated generic polygon labels, and that the resulting small dependency sums justified the local lemma. It continued scrutinizing this independence argument and the planar embedding step.

In the completed draft, it credited Golowich–Lin's multiplication-property construction and Dinur–Liu–Zhang's prescribed affine-line tests as inspiration for the polynomial pairings, while supplying its own proposed geometric construction [22, 15].

13 Repeated audits of a characteristic-two candidate

The assistant repeatedly challenged its own proposed characteristic-two counterexample to direct finiteness: elements $a, b \in K[G]$ with $ab = 1$ but $ba \neq 1$.

The candidate used a punctured affine space over $\mathbb{F}_q$, with q a power of two, covered regularly by full lines with incidence girth at least twelve. Polynomial interpolation at the holes produced vectors x_v whose outer products summed to zero on each surviving line and to $-I_m$ globally, where m counted holes. It credited the affine-line tests and multiplicative pairings to inspiration from Golowich and Lin's work on quantum LDPC codes and Dinur, Liu, and Zhang's codes on high dimensional expanders, without using a coding theorem as a dependency [22, 15]. Repeated audits concentrated on short-cycle counts, local degree-preserving replacements, and independence of random choices required by the Lovász local lemma.

The group construction glued copies of $\mathbb{F}_\ell^2$ along one-dimensional subgroups. Its proposed embedding proof converted a killed local element into a balanced planar graph. Degree and face bounds were intended to contradict Euler's formula; the assistant clarified the scope of its minimality argument:

VERBATIM EXCERPT

Minimal graph argument doesn't require planar balanced graph remains drawable as word diagram after contractions; minimal among all graphs satisfying combinatorial balances, not diagrams.

Subgroup averages then yielded rectangular matrices U, T with $UT = I_m$. Conjugations and character projections were intended to give a square one-sided inverse, while an embedded-subgroup specialization would make the reverse identity contradict a rank bound with $t > m$. Auxiliary matrix units supplied the proposed passage to scalar group-algebra elements. The assistant related this enlargement by a finite group factor to Dykema, Heister, and Juschenko's work on direct and stable finiteness [17].

The assistant used Linnik's prime bound [26] to choose $\ell \equiv -1 \pmod{q^{100}}$ with controlled size, keeping $\lfloor \log_q(\ell+1) \rfloor$ bounded as q grew. Attempts to replace this bound with elementary prime-divisor arguments, larger vector-space spreads, or explicit Mersenne primes ran into divisibility or interpolation-size obstacles, so it retained the theorem as a dependency. The assistant used canonical least choices to specify the finite objects in its construction.

On rereading the two papers of Ceccherini-Silberstein, Coornaert, and Phung, it retained surjunctivity as the hypothesis for their positive stable-finiteness results [12, 13].

The assistant ended by reopening its audit of the random-variable independence in the line construction and the reductions in the planar embedding argument; it had not identified a flaw in these checks.

14 Stress-testing a characteristic-two counterexample

The assistant examines its proposed counterexample to Kaplansky's direct-finiteness conjecture against algebraic and geometric obstructions. The goal is finite group-algebra sums a, b with $ab = 1$ but $ba \neq 1$. On rereading its draft, it retains the attribution of the finite-factor matrix reduction to Dykema, Heister, and Juschenko [17], and of the motivation for multiplicative polynomial pairings and prescribed affine-line tests to Golowich and Lin and to Dinur, Liu, and Zhang [22, 15]. The latter papers supply motivating themes rather than a theorem used in the candidate's proof.

The audit moves from possible external obstructions to the construction's geometric and algebraic joints. The assistant checks that the two surjunctivity results assume a surjunctive group or monoid, respectively [13, 12]. Looking at Farrell-Jones theory and $\text{CAT}(0)$ groups does not supply an unrestricted positive-characteristic theorem that rules out the candidate. The stable-finiteness discussion of Bartels, Lück, and Reich separates characteristic zero from the sofic case [6]; the latter uses Elek and Szabó's result [19]. A double-counting check clarifies why characteristic zero cannot support the proposed

weights: if every vertex has degree D and every line sum vanishes, then $D \sum_v W_v = 0$. Here D vanishes in the coefficient field.

VERBATIM EXCERPT

| Our mod2 D divisible q ensures. Thus no lift. Positive characteristic essential.

The assistant also considers whether the eigenvalue restrictions associated with Dodziuk, Linnell, Mathai, Schick, and Yates could obstruct an integral lift, but does not establish the required group-class hypothesis [16]. The geometry uses sparse deleted points, local line replacements preserving other degrees, and a finite local lemma to exclude short polygons. Conditional independence and witness counts receive particular scrutiny. Linnik's theorem supplies the prime-distribution bound used to keep the ambient dimension bounded as the field grows [26]; proposed elementary replacements encounter divisibility or interpolation difficulties.

Polynomial weights are intended to produce a nonzero global pairing despite zero line sums. A planar Euler-characteristic argument is then audited as the proposed certificate that the glued finite subgroups embed. Averaging projections yield rectangular matrices $UT = I_m$; packing them into square matrices would make the reverse identity imply a dimension- t identity factoring through dimension $m < t$. The assistant applies specialization only inside the smaller group algebra. Finite-group matrix units would convert the matrix obstruction into scalar sums.

The assistant next plans fuller justification of the geometric counts, embeddings, and algebraic reductions.

15 Auditing the proposed characteristic-two counterexample

The assistant scrutinized whether gluing finite groups preserved their embeddings. The target was finite sums $a, b \in K[G]$ in characteristic two with $ab = 1$ but $ba \neq 1$.

The assistant reread the supplied primary references and confirmed that their stable-finiteness results required surjectivity [12, 13].

The assistant credited multiplication properties in Golowich and Lin [22] and prescribed affine-line tests in Dinur, Liu, and Zhang [15] as motivation, without invoking their coding theorems. The construction used polynomial vectors x_v on a punctured affine space over $\mathbb{F}_q$, with $q = 2^h$. Proposed line trades preserved regularity, while independently chosen subspaces and local-lemma estimates excluded short incidence cycles. Interpolation was intended to give zero outer-product sums on every line and

$$\sum_v x_v x_v^\top = -I_m.$$

Averaging over embedded finite subgroups would then produce a rectangular factorization $UT = I_m$. Conjugating projections into orthogonal slots and finite-group matrix amplification were proposed to turn this into the scalar counterexample; the assistant related the amplification to Dykema, Heister, and Juschenko [17].

Linnik's theorem [26] remained an explicit dependency for choosing a suitably bounded prime $\ell \equiv -1 \pmod{q^{100}}$; attempted alternatives did not replace it. Possible residual-finiteness and curvature obstructions also remained unresolved.

VERBATIM EXCERPT

| Could N actually be trivial despite planar argument due to non-planarity of reduction? Need scrutinize signed wires again. If relators all local and group trivial, s reduces via conjugations in free group.

The embedding audit translated a supposedly killed generator into a planar weighted bipartite graph. Minimality was meant to force black degrees at least three, ordinary white degrees at least two, and all but possibly one face to have length at least twelve. These estimates would contradict Euler's formula. The revisions examined signed reductions, bridges, parallel edges, and neighbor merging, then made hole-deletion bounds, repeated-label independence, line parametrizations, and interpolation choices explicit. Writing B, W, E, F for black vertices, white vertices, edges, and faces, the assistant obtained $B \leq E/3$, $W \leq E/2 + 1/2$, and $F \leq E/6 + 1$, contradicting $B + W + F = E + 2$.

16 Testing the links in a proposed direct-finiteness counterexample

The recorded attempt repeatedly audited its proposed characteristic-two group-algebra elements satisfying $ab = 1$ but $ba \neq 1$. The aim shifted from reconstructing the entire construction to clarifying its subgroup embedding and projection packing. The minimization was over finite graphs satisfying the balance conditions, rather than only graphs obtained by particular reductions.

The assistant checked that the positive results of Ceccherini-Silberstein, Coornaert and Phung still required surjunctivity, under which group algebras over every field are stably finite [12, 13]. Its draft credited multiplication properties and prescribed affine-line tests in the coding papers of Golowich–Lin and Dinur–Liu–Zhang as inspiration for the polynomial pairing, without using a coding theorem as a dependency [22, 15].

The geometric proposal combined affine lines, deleted interpolation points, and random local replacements. Linnik's theorem controlled an odd prime ℓ [26], while a local-lemma argument was intended to ensure incidence girth at least twelve. Polynomial columns $x_v \in K^m$, for a finite field K of characteristic two, were designed to satisfy

$$\sum_{v \in f} x_v x_v^T = 0, \quad \sum_{v \in V} x_v x_v^T = -I_m.$$

The audit explained why summing line identities caused no immediate contradiction: the regular degree vanished in K .

Local groups $(\mathbb{Z}/\ell)^2$ were glued along cyclic subgroups. Their proposed injectivity depended on converting a collapsed generator into a planar balanced graph with one exceptional vertex. The assistant clarified its minimization argument by minimizing over all graphs satisfying the balance conditions, permitting contractions beyond a particular word reduction. Degree and face bounds were then intended to contradict Euler's formula.

Subgroup averages converted the polynomial identities into rectangular matrices with $UT = I_m$. Character projections packed them into square matrices. Assuming a reverse inverse would force I_t to factor through dimension $m < t$, after applying a homomorphism on an embedded base algebra. A finite auxiliary group supplied the proposed final matrix-to-scalar transfer. The draft related this enlargement to Dykema, Heister and Juschenko's result connecting direct finiteness after finite-group products with stable finiteness [17].

The assistant expanded the algebra and sharpened the planar argument, then examined the geometric existence argument and possible residual-finiteness obstructions.

17 Auditing a candidate through algebra and geometry

The assistant repeatedly challenged its proposed characteristic-two counterexample to direct finiteness: finite sums $a, b \in K[G]$ with $ab = 1$ but $ba \neq 1$. Its affine construction, developed within this attempt, combined a regular hypergraph of punctured lines, polynomial interpolation, and finite groups glued

along cyclic subgroups. The intended identities

$$\sum_{v \in f} x_v x_v^T = 0, \quad \sum_{v \in V} x_v x_v^T = -I_m$$

produced a proposed rectangular factorization $UT = I_m$; averaging idempotents, character twists, and matrix units were meant to convert it into the scalar counterexample.

With $q = 2^h$ and a prime $\ell \equiv -1 \pmod{q^{100}}$, the assistant used Linnik's bound to keep $\lfloor \log_q(\ell + 1) \rfloor$, and hence the ambient dimension, bounded as q grew [26]. The audit rechecked collision estimates, local line exchanges, short polygons, and a planar Euler argument intended to preserve the local subgroup embeddings. The assistant found that its attempted contradictions did not produce proofs. Finite global dimension and algebraic K -group information did not automatically provide positivity. Recalling its examination of Bartels, Lück, and Reich, the assistant noted that their argument used stable finiteness as a separate hypothesis, with the cited sufficient results requiring characteristic zero or soficity [6]; characteristic-zero lifting led to infinite support in a completion; approximate matchings did not give exact line partitions. An entropy objection also encountered jointly informative averages:

VERBATIM EXCERPT

| But collective conditional MI can be greater than sum due synergies (noises cancel via relations).

An optional increase of incidence girth from twelve to fourteen raised possible hyperbolicity and residual-finiteness consequences, prompting extra scrutiny. Six-sided configurations needed further occupancy controls, so this strengthening remained exploratory. The finite matrix block and finite-support reduction were clarified, while geometric classification remained unresolved.

The assistant's draft credited Dykema, Heister, and Juschenko for the finite-factor connection between direct and stable finiteness [17]. It also credited Golowich and Lin's work on products of algebraic codes and Dinur, Liu, and Zhang's work on high-dimensional expander codes with inspiring multiplicative pairings and prescribed affine-line tests; it treated these as inspiration, not coding-theory theorem dependencies [22, 15]. The two papers by Ceccherini-Silberstein, Coornaert, and Phung supplied the positive stable-finiteness results under a surjunctivity hypothesis [12, 13].

Wise's residual-finiteness theorem for acute-angled polygons of finite groups required at least four sides, so the assistant could not apply it directly to the triangular construction [39]. It ended this stage still seeking either a geometric classification or a contradiction.

18 Testing the candidate against geometric and algebraic obstructions

Across these consecutive audits, the assistant tried to break its proposed characteristic-two counterexample to direct finiteness: finite group-algebra sums satisfying $ab = 1$ but $ba \neq 1$.

The geometric audit rechecked sparse holes in an affine interpolation set, deletion estimates, and independence in a local-lemma argument excluding short polygons. Line replacements were intended to preserve regular degree while producing incidence girth at least twelve. Linnik's theorem remained necessary for the chosen parameter balance, keeping the ambient dimension bounded as the field size grew; elementary replacements stalled [26]. A planar balance-diagram argument was supposed to establish that local finite groups survived gluing, using degree bounds and Euler's formula.

Low-degree polynomial identities then supplied rectangular matrices U, T with $UT = I_m$. Character projections and conjugating generators were intended to turn this into a square-matrix obstruction: a reverse inverse would force I_t to have rank at most $m < t$. Finite-group matrix units would transfer

that obstruction to ordinary group-algebra elements. The assistant credited the finite-factor amplification theme to Dykema, Heister and Juschenko [17], and identified Golowich and Lin’s multiplication-property constructions and Dinur, Liu and Zhang’s prescribed affine-line tests as inspirations for its polynomial pairing [22, 15]; it did not invoke a coding theorem as a dependency.

Attention shifted toward structural reasons the candidate might be impossible. The assistant tested whether Berlai’s criterion for groups with a finitely generated residually finite normal subgroup and sofic quotient applied, but found no suitable normal subgroup [7]. It also considered Wise’s residual-finiteness result for acute polygons of finite groups with at least four sides, without establishing a reduction of its glued construction to that setting [39]. Its review recalled the residually amenable and sofic sufficient classes treated by Ara, O’Meara and Perera and by Elek and Szabó, respectively [2, 19]. The assistant distinguished lifting a projective-module class from lifting an actual module, and formal idempotent corrections from algebraic lifts with finite support. The assistant noted that the reviewed stable-finiteness results assumed surjunctivity [12, 13]. Neither those conditional results nor characteristic-zero positivity settled the characteristic-two case in its analysis. It also asked whether the glued group might be virtually special, sofic, or surjunctive, which would obstruct the proposed counterexample. Later it recalled Slofstra’s embedding theorem for solution groups and his nonclosure result for quantum correlations as reasons to question whether local commuting constraints must admit faithful finite-dimensional realizations [35, 36].

The investigation of the candidate’s compatibility with known structural finiteness results remained open.

19 Testing the candidate against finiteness and geometry

The recorded attempt repeatedly reconstructed its proposed characteristic-two counterexample to direct finiteness, the implication $ab = 1 \Rightarrow ba = 1$ in a group algebra. Its own construction used a regular hypergraph, vectors $x_v \in K^m$, and local groups of order $\kappa = \ell^2$, with ℓ odd. The intended identities were

$$\sum_v x_v x_v^\top = I_m, \quad \sum_{v \in f} x_v x_v^\top = 0, \quad m < t < m\kappa.$$

Subgroup averages were supposed to yield $UT = I_m$; character-based packing and a finite-group matrix block would convert this into scalar elements with unequal reverse product. Augmentation supplied the proposed rank obstruction. The draft connected its finite-group enlargement to the direct-to-stable-finiteness reduction of Dykema, Heister, and Juschenko [17]. It also acknowledged Golowich and Lin’s multiplication-property construction and Dinur, Liu, and Zhang’s prescribed affine-line tests as influences on its polynomial pairings, while using no coding-theory theorem as a proof dependency [22, 15].

The scrutiny shifted between algebra, geometry, and combinatorics. A finite quotient preserving every local group would contradict $m > t/\kappa$, but nonpositive curvature did not furnish such a quotient. Proper edge colorings were blocked by the polynomial identities. Lifting those identities to characteristic zero also failed: summing over lines multiplies the global pairing by the degree, which vanishes only in the chosen characteristic. The assistant used Bartels, Lück, and Reich’s discussion of stable finiteness and projective-module classes to check whether Farrell–Jones theory supplied the missing positivity; it found that the cited argument retained a separate stable-finiteness assumption [6].

VERBATIM EXCERPT

| FJ computes $K0$ as abelian group with action but not monoid positivity.

The assistant later returned to reconstructing the candidate from its abstract identities.

The assistant checked stable-finiteness theorems of Ceccherini-Silberstein, Coornaert, and Phung, whose conclusions retained a surjunctivity hypothesis [12, 13]. Rechecking planar embedding arguments and local-lemma estimates found no decisive defect; a stronger exclusion of six-step polygons remained exploratory. Linnik’s theorem remained a substantial dependency for polynomially bounded primes in the required congruence classes after prime-power replacements failed [26].

The assistant retained the candidate after these checks and continued testing structural obstructions. Its proposed exclusion of six-step polygons still required careful counting of repeated labels and exceptional-line witnesses. In considering whether a stronger girth bound would force a familiar geometric class, the assistant recalled Ollivier and Wise’s work on cubulating random groups and on graphical small cancellation with Kazhdan subgroups. It treated these as comparisons and did not establish that either result applied to its own group [29, 30].

20 Repeated audits and an unused elementary alternative

The assistant repeatedly tested its proposed characteristic-two counterexample to direct finiteness against possible combinatorial, representation-theoretic, and lifting obstructions.

The candidate combined a regular hypergraph of affine lines, polynomial interpolation at removed points, and glued finite subgroups. On rereading its construction, the assistant retained its attribution of the low-degree line tests and multiplicative pairings to themes in the coding work of Golowich and Lin and of Dinur, Liu, and Zhang [22, 15]. If m points were removed and $t > m$ remained, interpolation columns were intended to satisfy

$$\sum_v x_v x_v^T = -I_m, \quad \sum_{v \in f} x_v x_v^T = 0.$$

The assistant used these identities to rule out perfect matchings: the uncovered vertices must supply a rank- m sum of rank-one matrices, requiring at least m uncovered vertices. In its construction, subgroup averages yielded $UT = I_m$; compression and character specialization would make a reverse identity imply the impossible rank inequality $t \leq m$. The subgroup-embedding argument was rechecked through planar degree and face bounds contradicting Euler’s formula. The construction also retained its connection between a finite auxiliary group and matrix stable finiteness, attributed to Dykema, Heister, and Juschenko [17]; the model-theoretic reference recalled Dykema and Juschenko’s precise criterion using all finite direct factors [18].

Attempts to recover characteristic-zero positivity failed to preserve either finite support or real trace equality:

VERBATIM EXCERPT

Trace is predetermined rational by p-adic cyclic calculation and formal terms, maybe real trace differs even with simultaneous convergence; p-adic equality in infinite rational sums doesn't force real equality.

The assistant rechecked that the stable-finiteness results it had consulted required surjunctivity [12, 13]. Its checks of finite quotients, random-subspace independence, short polygons, and matching heuristics revealed no contradiction. It also revisited its comparison with Wise’s residual-finiteness theorem for acute-angled polygons of finite groups with at least four sides, whose scope did not cover the triangular complexes it was considering [39].

A subsequent detour sought to replace Linnik’s theorem [26], which supplied the original prime parameters. Chebyshev’s prime-counting bound and a Rankin smooth-number estimate suggested varying the coefficient characteristic and enlarging the local finite fields. The assistant estimated that the

probabilistic construction might survive the resulting growing dimension, but retained the original dependency.

The assistant nevertheless returned to doubts about hidden global constraints, asking whether the affine geometry and degree forced short cycles that its estimates had missed. It began extending the polygon analysis to length six, but left that extension unfinished and retained the original prime-parameter construction.

21 Testing the limits of a characteristic-two construction

The assistant repeatedly challenged its proposed failure of direct finiteness, the implication $ab = 1 \Rightarrow ba = 1$ for group-algebra elements.

First, refined counts of six-step polygons suggested strengthening an affine-line incidence graph from girth twelve to fourteen. Combined with triangle angles $\pi/7, \pi/3, \pi/2$, this raised the speculative possibility of a hyperbolic example. The subsequent reconstruction retained Linnik's theorem [26] for choosing an odd prime ℓ with $D = \ell + 1$ divisible by a large power of the field order. Random subspaces, sparse holes, and local line replacements were intended to produce a regular hypergraph with no short incidence cycles. The assistant continued using low-degree line tests and multiplicative pairings motivated by Golowich–Lin and Dinur–Liu–Zhang [22, 15], without invoking a coding-theory theorem. Polynomial interpolation supplied vectors $x_v \in K^m$, over characteristic two, satisfying

$$\sum_{v \in f} x_v x_v^T = 0, \quad \sum_v x_v x_v^T = -I_m.$$

Subgroup-averaging idempotents were then meant to yield $UT = I_m$. Augmentation would obstruct the reverse identity through a rank comparison; an auxiliary finite-group matrix block would convert the proposed matrix example into scalar elements, following the finite-factor connection discussed by Dykema–Heister–Juschenko [17].

The assistant found no contradiction in its tests using finite quotients, coding theory, matchings, and algebraic K -theory. It revisited Wise's residual-finiteness theorem for acute-angled negatively curved polygons of finite groups with at least four sides [39], but found no application to its triangular construction. Its comparison with Bartels–Lück–Reich [6] distinguished additive projective classes from positive dimensions: their stable-finiteness inputs had characteristic-zero or sofic hypotheses, so the assistant did not derive the required conclusion from Farrell–Jones assembly alone. Summing the hyperedge identities also exposed why the same weights could not simply be lifted to characteristic zero:

VERBATIM EXCERPT

Thus characteristic- p identity inherently due $D=0$. p -adic lift must alter supports adding higher-order terms from nonlocal group elements; trace contradiction possible only if accidentally still finite.

Rereading the surjectivity results [12, 13], the assistant confirmed that their stable-finiteness conclusions required surjectivity hypotheses. The remaining audit returned to whether local groups survived their gluing: planar degree and face-length bounds were compared with Euler's formula, with particular attention to bridges and repeated boundaries. The assistant retained the proposal and continued scrutinizing whether the local groups survived the gluing.

22 Repeated scrutiny of the proposed counterexample

The closing stages tested the assistant's proposed characteristic-two counterexample to direct finiteness: finite group-algebra sums satisfying $ab = 1$ but $ba \neq 1$.

The construction combined a punctured affine space over $\mathbb{F}_q$, low-degree interpolation, and finite local groups $K_v \cong (\mathbb{Z}/\ell)^2$ glued along cyclic line subgroups. Prime selection used $\ell \equiv -1 \pmod{q^{100}}$ and continued to depend on Linnik's theorem [26]. A proposed planar-diagram argument used incidence girth twelve and degree bounds to contradict Euler's formula, thereby defending the local groups' embeddings. Averaging idempotents then entered rectangular matrices with $UT = I_m$. Conjugating elements and character projections were intended to produce square matrices whose reverse identity would force an impossible rank- m factorization of I_t , where $t > m$. A finite group-algebra block would convert this matrix obstruction into individual elements. The write-up connected this finite-factor step to Dykema, Heister, and Juschenko's work on direct and stable finiteness [17]. It also credited multiplication properties and prescribed affine-line tests in the coding work of Golowich and Lin and of Dinur, Liu, and Zhang as motivation for its polynomial pairing, while using no coding theorem as a premise [22, 15].

The assistant rechecked the combinatorial probability estimates, interpolation, embeddings, and algebraic packing. It examined repeated polygon labels, distinct rectangle witnesses, and the number of holes left after deleting obstructions. Simple retractions available for graph products failed to apply because of the local linear dependencies; it identified no applicable residual-finiteness theorem. The cited stable-finiteness theorems retained their surjunctivity hypothesis [12, 13], so the assistant found that they did not settle the unrestricted target.

Integer lifts suggested a self-adjoint operator with a nonintegral rational spectral atom $-1/2$. The assistant recalled the algebraic-eigenvalue work of Dodziuk, Linnell, Mathai, Schick, and Yates while exploring possible spectral obstructions [16]. It found no universal exclusion applicable to its construction. Positivity and integral moments did not suffice, and the potentially large spectral range frustrated the suggested polynomial-approximation argument. Complex unitary approximation likewise supplied no justified transfer to characteristic two.

One correction concerned a speculative extension excluding six-step polygons: tenfold subspace independence was insufficient before repeated labels had been established, since twelve subspaces could occur.

VERBATIM EXCERPT

Without direct for all six G, E labels may not repeat! For $k \leq 5$ count ≤ 10 . $k=6$ not covered by Vandermonde in $n=10b$.

Increasing the ambient dimension from $10b$ to $12b$ was proposed, followed by reconsideration of the counting bounds. In testing this extension, it compared the proposed triangular complex with Wise's residual-finiteness results for negatively curved polygons of finite groups [39] and with his cubulation results for small-cancellation groups [40]. It questioned whether the required polygon or wall conditions held. It also recalled Wise's work on positive one-relator groups as another possible route, without establishing applicability [41]. The assistant left its suggested negative-curvature, residual-finiteness, and determinant consequences unexplored in its final argument.

Part II: Gottschalk surjunctivity from one-sided inverses

Original prompt (excerpts)

Problem

Let G be any group and let A be a nonempty finite set. Write A^G for all functions $x: G \rightarrow A$. A cellular automaton with finite memory is a map $\tau: A^G \rightarrow A^G$ for which there exist a finite subset M of G and a function $\phi: A^M \rightarrow A$ such that

$\tau(x)(g) = \phi((x(gm))_{m \in M})$
 for every x in A^G and g in G . The same local function ϕ is used at every g .

Prove or disprove that every such injective map τ is surjective, for every choice of G, A, M, ϕ .
 Injective means $\tau(x) = \tau(y)$ implies $x = y$; surjective means every y in A^G has the form $\tau(x)$.
 There is no assumption that G is finite, amenable, residually finite, sofic, or finitely presented.
 A counterexample must give an allowed group and a finite-memory uniform local rule that is injective but not surjective.

23 From one-sided inverses to a cellular automaton

The companion investigation asked whether every injective uniform cellular automaton on a full shift over a finite alphabet must be surjective. Its starting context distinguished this unrestricted question from positive results for sofic groups and from results that already assumed surjunctivity. Phung's work on non-uniform cellular automata supplied that contextual distinction [32]. A previous characteristic-two argument asserted the existence of a finite field K , a group G , and finite sums $a, b \in K[G]$ with $ab = 1$ and $ba \neq 1$. The assistant took that assertion as its input and sought the actual local rule, rather than stopping at the statement that the group was nonsurjunctive.

The previous characteristic-two argument also referred to Elek and Szabó's sofic-group theorem [19], the group and monoid surjunctivity theorems of Ceccherini-Silberstein, Coornaert, and Phung [13, 12], and the finite-factor reduction of Dykema, Heister, and Juschenko [17]. The prescription used Linnik's prime bound, cited there through Xylouris [42], while the polynomial construction credited Golowich–Lin and Dinur–Liu–Zhang as inspiration [22, 15]. These references entered through that argument; the companion conversion retained its single input of finite-field elements with unequal one-sided products.

The conversion took $A = K$ and associated to every finite sum $c = \sum_u c_u u$ the map

$$T_c(x)(g) = \sum_{u \in G} c_u x(gu), \quad x \in K^G.$$

Finite support of c makes the sum finite even when x is an arbitrary configuration. For $\tau = T_b$, the memory is $M = \text{supp}(b)$ and the uniform rule is $\phi(p) = \sum_{u \in M} b_u p(u)$. The same coefficients apply at every g , and finiteness of K makes this a finite alphabet.

The assistant immediately checked the side on which multiplication appeared:

VERBATIM EXCERPT

Correct order? $T_c(x)(g) = \sum c_m x(gm)$. $(T_a(T_b(x)))(g) = \sum a_h b_k x(ghk)$. $T_b T_a = T_{\{ba\}}$. So yes.

Grouping the finite double sum by uv gives $T_c \circ T_d = T_{cd}$. Thus $T_a \circ T_b = T_1 = \text{id}$, and T_b is injective. The assistant checked the order because G need not be commutative, and chose b for the desired automaton.

To obtain nonsurjectivity, the assistant used the nonzero coefficient of $ba - 1$. Choose h with $(ba)_h \neq (1)_h$ and let $y = \delta_h$ be the configuration equal to 1_K at h and zero elsewhere. Every configuration in the image of T_b is fixed by $T_b T_a = T_{ba}$, whereas

$$T_{ba}(y)(e) = (ba)_h \neq (1)_h = y(e).$$

Consequently this particular y has no preimage. The argument uses the full shift, not just finitely supported inputs; the delta configuration serves only as the witness outside the image.

The assistant returned repeatedly to the distinction between this implication and an equivalence of the two conjectural targets:

VERBATIM EXCERPT

| Our use is not identifying equivalence; direct finite failure gives one counterexample, no reverse.

It also considered replacing the witness group by a familiar ambient group, such as $\text{Sym}(\mathbb{N})$, through an embedding of the countable group. It kept the original group and its finite sums instead. The final write-up specified the characteristic-two theorem and its prescribed witnesses, gave the local rule above, and concluded with the explicit configuration outside its image. The companion argument's mathematical dependency remained the supplied one-sided-inverse theorem; it did not reconstruct the affine incidence construction that produced a and b .

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